An improved bound for the exceptional set of the Erdős–Straus equation

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Abstract

Let $E(N)$ be the number of integers $2 \le n \le N$ for which

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

has no solution in positive integers. Vaughan proved $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$. We prove

$$E(N) \ll N \exp\{-c(\log N)^{3/4}\}.$$

For each $a \ge 1$, the divisors of a^2 supply congruence classes modulo $4a-1$ on which such representations exist. For a large parameter X, we select a family whose reciprocal moduli sum to order $(\log X)^3$. Averaging over small divisors retains this weight while controlling shared prime factors. A finite counting argument gives the stated bound at the scale $\log X \asymp (\log N)^{1/4}$. The same argument gives this exponent for every fixed numerator $m \ge 4$.

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1 Introduction and main theorem

The Erdős–Straus conjecture asserts that for every integer $n \ge 2$ the equation

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers. Let $E(N)$ count the exceptions $2 \le n \le N$. Vaughan proved the classical estimate

$$E(N) \ll N \exp\{-c(\log N)^{2/3}\}$$

for some absolute $c > 0$ [15, p. 193, (2)]. Our main result improves the exponent.

Theorem 1 (main theorem)**.** *There is an absolute constant $c > 0$ such that, for all $N \ge 3$,*

$$E(N) \ll N \exp\{-c(\log N)^{3/4}\}.$$

The same saving holds relative to the primes: the number of exceptional primes up to N is $\ll \pi(N) \exp\{-c(\log N)^{3/4}\}$ after decreasing c.

For $m \ge 4$, let $E_m(N)$ count the positive integers $n \le N$ for which m/n is not a sum of three unit fractions.

Theorem 2 (Erdős–Straus–Schinzel exceptional sets for a fixed numerator)**.** *For each fixed integer $m \geq 4$, there are constants $c_m, C_m > 0$ such that, for all $N \geq 3$,*

$$E_m(N) \leq C_m N \exp\{-c_m (\log N)^{3/4}\}.$$

The cases $m = 2, 3$ have no exceptions, since

$$\frac{2}{n} = \frac{1}{n} + \frac{1}{2n} + \frac{1}{2n}, \qquad \frac{3}{n} = \frac{1}{n} + \frac{1}{n} + \frac{1}{n}.$$

Pomerance–Weingartner [14, Thm. 1.3] prove $E_m(N) \leq N \exp\{-C(\log N)^{2/3}/\varphi(m)^{1/3}\}$ uniformly for $4 \leq m \leq (\log N)^2$. For each fixed m, Theorem 2 improves the exponent from $2/3$ to $3/4$, with constants depending on m.

Example 1.1. For $n = 2521$, take $a = 12$ and $h = 16$. Then $h \mid a^2$ and

$$4a - 1 = 47, \qquad n + 4h = 2585 = 47 \cdot 55.$$

Lemma 2.3 gives

$$\frac{4}{2521} = \frac{1}{644} + \frac{1}{30252} + \frac{1}{1217643}.$$

We use the Type II congruences

$$4a - 1 \mid n + 4h, \qquad h \mid a^2.$$

The choices $h = 1, a, a^2$ give the shifts $-4, -1, -a$ modulo $4a - 1$, respectively. Classical congruence criteria appear in Yamamoto [18, Lemmas 1–2]. The sum over the remaining divisors has cubic logarithmic order; see Proposition 4.2.

In the selected family, compatible congruences with the same large prime coincide. We bound the remaining overlaps by sums over common small factors. Two divisor sums give weight $(\log X)^2$; averaging over small divisors supplies $\log z$, which combines with the rough cofactor sum $\log X/\log z$ to give the third logarithm.

Vaughan's sieve estimates [16, Cor. 3.1.1] and Elsholtz's treatment of sums of k unit fractions [5, Thm. 1.3 and Sec. 5] provide earlier approaches to exceptional sets; the latter explicitly controls distinctness of the eliminated residue classes. Elsholtz and Tao proved bounds for the average number of representations over primes [7], and Elsholtz–Planitzer give further results on the number of representations [6]; Bloom–Elsholtz survey Egyptian fractions [2]. Bright and Loughran studied the associated surfaces through the Brauer–Manin obstruction [4].

Bello-Hernández, Benito, and Fernández [1] study a divisor parametrization with prescribed common divisibility and compare it with Type I/II descriptions.

Remark 1.2 (the exponent $3/4$). For a family of congruences with total weight $\mu \asymp (\log X)^k$ and the overlap estimates below, the level condition of Theorem 5.4 has the form $\log P(z) + O(\mu \log X) \leq \log N$. When $\log P(z) = O((\log X)^{k+1})$, choosing $\log X$ of order $(\log N)^{1/(k+1)}$ gives a saving $\exp\{-c(\log N)^{k/(k+1)}\}$. The family with prime moduli used in the Vaughan method has quadratic mass; see Pomerance–Weingartner [14, Lemma 4.1]. Here the cubic mass gives $3/4$.

Formalization. The companion Lean 4 development [12] proves Theorems 1 and 2, the bound for exceptional primes, Example 1.1, and every numbered result of Sections 2–6, using only Mathlib and Lean's standard axioms. Its proof of Theorem 3.7 adapts an existing open source formalization of the Bombieri–Vinogradov theorem, credited in the repository.

2 Congruences

Fix $m \geq 4$. A Type II representation for a prime denominator is one in which exactly two denominators are divisible by the prime. The following is a reformulation of the criteria of Elsholtz–Tao [7, Sec. 2] and Pomerance–Weingartner [14, Prop. 2.3 and Cor. 2.4]. The proof of Theorem 1 uses only Lemma 2.3.

Theorem 2.1 (Type II representations)**.** *Let p be a prime. Suppose that positive integers d, f, A, B satisfy*

$$(mdfA - 1)B = Ap + d. \tag{2}$$

Then

$$\frac{m}{p} = \frac{1}{pdfA} + \frac{1}{pfAB} + \frac{1}{dfB}.$$

Conversely, every Type II representation can be put in this form after permuting the two denominators divisible by p.

Proof. Putting the three displayed fractions over the common denominator $pdfAB$ gives numerator $B+d+Ap = mdfAB$ by (2). The third denominator cannot also be divisible by p, since then the three fractions would sum to at most $3/p$. Conversely, write a Type II solution as

$$\frac{m}{p} = \frac{1}{pu} + \frac{1}{pv} + \frac{1}{w}, \qquad p \nmid w.$$

Let $g = (u, v)$ and write $u = gd$, $v = gB$, where $(d, B) = 1$. Clearing denominators and dividing by g gives

$$mgdBw = w(d + B) + pgdB.$$

Modulo d, this gives $wB \equiv 0 \pmod{d}$, so $d \mid w$ because $(d, B) = 1$. Modulo B, the same argument gives $B \mid w$. Thus $dB \mid w$; write $w = dfB$ with $f \geq 1$. Substitution and division by dB yield

$$mgdfB = f(d + B) + pg.$$

Thus $f \mid pg$. Since $p \nmid w$, we have $(f, p) = 1$, so $f \mid g$; write $g = fA$. Dividing the last identity by f now gives

$$mdfAB = d + B + pA,$$

which is (2). The formulas $u = dfA$, $v = fAB$, and $w = dfB$ follow from the definitions. For $m = 4$ this is equivalent to the Elsholtz–Tao Type II normal form [7, Prop. 2.6, (2.18), (2.20), (2.22)]. □

Put $a = dfA$, $h = d^2f$, and $q_a = ma - 1$. Then $h \mid a^2$.

Theorem 2.2 (Type II congruences)**.** *Let p be a prime and $a \geq 1$. There is a Type II solution with first denominator pa if and only if there exists a divisor $h \mid a^2$ such that*

$$ma - 1 \mid p + mh. \tag{3}$$

When (3) *holds, write $h = d^2f$ and $a = dfA$. Then*

$$B = \frac{Ap + d}{mdfA - 1}$$

is integral, and Theorem 2.1 gives a Type II representation.

Proof. For a representation in Theorem 2.1, the identity

$$A(p + md^2f) = Ap + d + d(mdfA - 1)$$

shows that $ma - 1$ divides $A(p + mh)$. Since $(A, ma - 1) = 1$, it divides $p + mh$.

Conversely, for $h \mid a^2$ put

$$g = (a, h), \qquad d = h/g, \qquad A = a/g.$$

Then $(d, A) = 1$, and $h \mid a^2$ gives $d \mid gA^2$. Hence $d \mid g$; set $f = g/d = g^2/h$. Now $g = df$, so $a = dfA$ and $h = d^2f$. Conversely, any such decomposition with $(d, A) = 1$ has $(a, h) = df$, which recovers d, f, A from these formulas and proves uniqueness.

Finally,

$$Ap + d = A(p + mh) - d(ma - 1).$$

The congruence (3) therefore makes $B = (Ap + d)/(ma - 1)$ integral. Both numerator and denominator are positive, so Theorem 2.1 applies. □

For arbitrary positive denominators we use the following sufficient condition.

Lemma 2.3 (a sufficient condition)**.** *Let $n \geq 1$, let $a \geq 1$, let $q = ma - 1$, and let e be a positive divisor of a^2. If*

$$n \equiv -me \pmod{q},$$

then m/n is a sum of three positive unit fractions.

Proof. Put $A = an$. Since $ma \equiv 1 \pmod{q}$, the congruence on n gives $A \equiv -e \pmod{q}$. Since $e \mid a^2$ and $(a, q) = 1$, also $(e, q) = 1$. Define

$$x = A, \qquad y = \frac{A + e}{q}, \qquad z = \frac{A + A^2/e}{q}.$$

Since $e \mid a^2$, the quantity $A^2/e = (a^2/e)n^2$ is an integer. The congruence $A \equiv -e \pmod{q}$ makes y integral; squaring and multiplying by the inverse of e modulo q gives $A^2/e \equiv e \pmod{q}$, so z is integral too. Both are positive. Moreover

$$qy - A = e, \qquad qz - A = A^2/e,$$

and hence $(qy - A)(qz - A) = A^2$. Expanding and cancelling A^2 gives

$$q^2yz - qA(y + z) = 0.$$

Dividing by qyz yields $q = A(1/y + 1/z)$. Therefore

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{1 + q}{an} = \frac{m}{n}.$$

□

3 Preliminary estimates

Fix

$$(\eta, \sigma, \lambda, \theta, \beta, \nu) = \left(\frac{1}{100}, \frac{1}{20}, \frac{1}{10}, \frac{3}{10}, \frac{2}{3}, \frac{8}{25}\right). \tag{4}$$

Thus $3\eta < \lambda$, $\eta + \sigma < \lambda$, $\lambda + \eta < \theta$, and $\theta < \nu < \beta/2 < \beta < 1 - \sigma$. Put

$$z = (\log X)^4, \qquad U = (\log X)^8, \qquad P(z) = \prod_{\ell \leq z} \ell.$$

The product is over primes. Implied constants and thresholds depend at most on the fixed numerator m and on parameters declared fixed. Uniformity is with respect to all other variables in the stated ranges.

The family in Section 6 is constructed as follows. Choose $e = rs^2$ with r, s squarefree and coprime, a small divisor $d_- \mid P(z)$, a squarefree cofactor d_+ with $(d_+, P(z)) = 1$, and a prime p. Require $d_-d_+p \equiv -1 \pmod{m \operatorname{rad}(e)}$. Then $a = (d_-d_+p + 1)/m$ satisfies $e \mid a^2$, and Lemma 2.3 supplies the class $n \equiv -me \pmod{d_-d_+p}$. After fixing $n \equiv b \pmod{P(z)}$, require $b \equiv -me \pmod{d_-}$; the remaining modulus is d_+p. Section 5 gives the counting argument for these classes.

We use Abel summation in the following finite form. If $A(t) = \sum_{n \leq t} a_n$ and f is continuously differentiable on $[u, v]$, then

$$\sum_{u < n \leq v} a_n f(n) = A(v)f(v) - A(u)f(u) - \int_u^v A(t)f'(t)\,dt.$$

This follows by writing $f(n) = f(v) - \int_n^v f'(t)\,dt$ and interchanging the finite sum and integral.

Lemma 3.1 (prime sums)**.** *For $x \geq 2$,*

$$\sum_{p \leq x} \log p \ll x, \qquad \sum_{p \leq x} \frac{\log p}{p} = \log x + O(1), \qquad \sum_{p \leq x} \frac{1}{p} = \log\log x + B + O(1/\log x).$$

Moreover,

$$\prod_{p \leq x} (1 - p^{-1})^{-1} \asymp \log x, \qquad \prod_{z < p \leq x} (1 + p^{-1}) \asymp \frac{\log x}{\log z} \quad (2 \leq z \leq x).$$

In particular, for fixed $0 < a < b$,

$$\sum_{X^a < p \leq X^b} \frac{1}{p} = \log(b/a) + O_{a,b}(1/\log X).$$

Proof. We recall the argument; see also [13, Chaps. 2–3]. Put $P_*(n) = \prod_{p \leq n} p$. Strong induction gives $P_*(n) \leq 4^n$. For even $n > 2$, $P_*(n) = P_*(n-1)$. For $n = 2k+1$, each prime $k+1 < p \leq 2k+1$ divides $\binom{2k+1}{k}$, and the two equal central coefficients give $\binom{2k+1}{k} \leq 4^k$. Hence $P_*(2k+1) \leq P_*(k+1)4^k \leq 4^{2k+1}$. Taking logarithms proves the first estimate.

Let $\Lambda(p^j) = \log p$ and let $\Lambda(n) = 0$ otherwise. With $\psi(x) = \sum_{n \leq x} \Lambda(n)$, the first estimate gives

$$\psi(x) = \sum_{j \leq \log x / \log 2} \sum_{p \leq x^{1/j}} \log p \ll x + \sqrt{x} \log x \ll x.$$

Counting prime factors in $n!$ gives Legendre's identity and therefore

$$\log(n!) = \sum_{d \leq n} \Lambda(d) \lfloor n/d \rfloor, \qquad 0 \leq n \sum_{d \leq n} \frac{\Lambda(d)}{d} - \log(n!) \leq \psi(n) \ll n.$$

The integral bounds for $\sum_{j \leq n} \log j$ give $\log(n!) = n \log n + O(n)$. The proper prime powers contribute a bounded amount, since

$$\sum_{p} \sum_{j \geq 2} \frac{\log p}{p^j} = \sum_{p} \frac{\log p}{p(p-1)} \leq \sum_{n \geq 2} \frac{\log n}{n(n-1)} < \infty.$$

Thus $B_1(x) := \sum_{p \leq x} (\log p)/p = \log x + O(1)$, first for integer and then for real x. Abel summation gives

$$\sum_{p \leq x} \frac{1}{p} = \frac{B_1(x)}{\log x} + \int_2^x \frac{B_1(t)}{t(\log t)^2}\, dt.$$

The error integral converges, with tail $O(1/\log x)$, proving the third estimate and the formula on a fixed prime window.

Finally, $-\log(1 - p^{-1}) = p^{-1} + O(p^{-2})$ and $\sum_p p^{-2} < \infty$ give the first product estimate. The identity $1 + p^{-1} = (1 - p^{-2})/(1 - p^{-1})$ gives the second: the product of the factors $1 - p^{-2}$ is bounded away from zero, and the quotient of the first product estimates is $\asymp \log x / \log z$.

For the lower bound on the prime counting function used in the introduction, Legendre's formula also gives

$$\frac{4^n}{2n+1} \leq \binom{2n}{n} \leq e^{\psi(2n)}.$$

Indeed, each exponent in the binomial coefficient is at most the number of powers of that prime not exceeding $2n$. Hence $\psi(x) \gg x$. Since $\psi(x) - \sum_{p \leq x} \log p \ll \sqrt{x} \log x = o(x)$, it follows that $\pi(x) \gg x/\log x$. □

Put $\gamma(k) = \mu^2(k)/\varphi(k)$ and $C_\gamma = \sum_{k \geq 1} \gamma(k)/k = \prod_p (1 + 1/(p(p-1))) < \infty$. The identity $n/\varphi(n) = \sum_{k \mid n} \gamma(k)$ and positive summation give

$$\sum_{n \leq T} \frac{n}{\varphi(n)} \ll T, \quad \sum_{n \leq T} \frac{1}{\varphi(n)} \ll \log(2T), \quad \sum_{k \leq T} \gamma(k) \ll \log(2T), \quad \sum_{k > T} \frac{\gamma(k)}{k} \ll \frac{\log(2T)}{T}. \tag{5}$$

In particular $M/\varphi(M) \ll \log(2M)$.

Lemma 3.2 (ordinary squarefree reciprocal sums)**.** *Let $0 < a_0 < b_0$ and $c_0 > 0$ be fixed. Uniformly for $X^{a_0} \le U_0 < U_1 \le X^{b_0}$ with $U_1/U_0 \ge X^{c_0}$, one has*

$$\sum_{\substack{U_0 < n \le U_1 \\ n \text{ squarefree}}} \frac{1}{n} \asymp_{a_0,b_0,c_0} \log \frac{U_1}{U_0}.$$

If M is squarefree and $M \le X^{C_0}$ for a fixed C_0, then

$$\sum_{\substack{U_0 < n \le U_1 \\ n \text{ squarefree} \\ (n,M)=1}} \frac{1}{n} \asymp_{a_0,b_0,c_0,C_0} \frac{\varphi(M)}{M} \log \frac{U_1}{U_0}. \tag{6}$$

For every positive integer D, every class $b \pmod D$, and every $1 \le U_0 < U_1$,

$$\sum_{\substack{U_0 < n \le U_1 \\ n \equiv b \pmod D}} \frac{1}{n} \ll \frac{1}{D} \log \frac{U_1}{U_0} + \frac{1}{U_0}. \tag{7}$$

Proof. For positive integers M, d with M squarefree and $(M,d) = (b,d) = 1$, and for $x \ge 1$, expand $\mu^2(n) = \sum_{k^2 \mid n} \mu(k)$. Only $(k, Md) = 1$ contributes. Writing $n = k^2 u$, inclusion–exclusion over the divisors of M gives

$$\#\{u \le x/k^2 : u \equiv bk^{-2} \pmod d,\ (u,M) = 1\} = \frac{\varphi(M)}{Md}\frac{x}{k^2} + O(\tau(M)),$$

because each divisor $t \mid M$ imposes one class modulo dt. Summing over $k \le \sqrt{x}$ and completing the series of reciprocal squares yields, with an absolute implied constant,

$$\#\{n \le x : n \equiv b \pmod d,\ (n,M) = 1,\ n \text{ squarefree}\} = \kappa(M,d)\frac{\varphi(M)}{Md}x + O(\tau(M)\sqrt{x}),$$
$$\kappa(M,d) = \prod_{p \nmid Md} (1 - p^{-2}) \in [1/\zeta(2), 1].$$

The omitted series tail is $O(x^{-1/2})$, with an absolute constant. For $1 \le A < B$, partial summation therefore gives, with an absolute implied constant,

$$\sum_{\substack{A < n \le B \\ n \equiv b \pmod d \\ (n,M)=1}} \frac{\mu^2(n)}{n} = \kappa(M,d)\frac{\varphi(M)}{Md} \log \frac{B}{A} + O(\tau(M)/\sqrt{A}).$$

For squarefree M, $\tau(M) = \prod_{p \mid M} 2 \ll_\varepsilon M^\varepsilon$: the primes below $2^{1/\varepsilon}$ contribute a bounded product, and each remaining factor satisfies $2 \le p^\varepsilon$. Choosing ε sufficiently small gives $\tau(M) \ll_{a_0,C_0} X^{a_0/4}$ for $M \le X^{C_0}$. Take $d = 1$, $A = U_0$, and $B = U_1$. The error is $O(X^{-a_0/4})$, whereas $\varphi(M)\log(U_1/U_0)/M \gg_{c_0,C_0} 1$ by (5). This proves (6); setting $M = 1$ gives the first assertion. For an arithmetic progression of spacing D, charge its first term to $1/U_0$ and each subsequent $1/n$ to $D^{-1}\int_{n-D}^{n} dt/t$. This proves (7). □

Lemma 3.3 (squarefree weighted sums)**.** *For every fixed positive integer M, as $T \to \infty$,*

$$\sum_{\substack{s \le T \\ s \text{ squarefree} \\ (s,M)=1}} \frac{\varphi(s)}{s^2} \asymp_M \log T, \qquad \sum_{\substack{s \le T \\ s \text{ squarefree} \\ (s,M)=1}} \frac{1}{s} \asymp_M \log T.$$

Proof. Replace M by $\operatorname{rad}(M)$. Lemma 3.2 on $(\sqrt{T}, T]$ gives the second lower bound, and the harmonic sum gives its upper bound. Cauchy–Schwarz and (5) give

$$\left(\sum_{\substack{s\le T\\ s\text{ squarefree}\\ (s,M)=1}}\frac{1}{s}\right)^2 \le \left(\sum_{\substack{s\le T\\ s\text{ squarefree}\\ (s,M)=1}}\frac{\varphi(s)}{s^2}\right)\left(\sum_{s\le T}\frac{1}{\varphi(s)}\right) \ll \log T \sum_{\substack{s\le T\\ s\text{ squarefree}\\ (s,M)=1}}\frac{\varphi(s)}{s^2}.$$

This proves the first lower bound; its upper bound follows from $\varphi(s) \le s$. □

Lemma 3.4 (rough reciprocal sums over an initial segment)**.** *Let $0 < a_0 \le b_0$ and $C_0 > 0$ be fixed. Uniformly for $X^{a_0} \le x \le X^{b_0}$ and $1 \le M \le X^{C_0}$,*

$$\sum_{\substack{n\le x\\ n\text{ squarefree}\\ (n,P(z))=1\\ (n,M)=1}}\frac{1}{n} \asymp_{a_0,b_0,C_0} \frac{\log X}{\log z}.$$

Proof. Put

$$H_B(x) = \sum_{\substack{n\le x\\ n\text{ squarefree}\\ (n,B)=1}}\frac{1}{n}.$$

The unique decomposition $n = ab^2$, with a squarefree, gives $\sum_{n\le x} n^{-1} \le H_1(x)\sum_{b\ge 1} b^{-2} \le 2H_1(x)$. For squarefree B, the decomposition $n = gt$, where $g = (n, B)$, gives

$$H_1(x) = \sum_{g\mid B}\frac{1}{g}H_B(x/g) \le H_B(x)\prod_{p\mid B}(1+p^{-1}).$$

Take $B = \operatorname{rad}(P(z)M)$. Mertens' theorem gives $\prod_{p\le z}(1+p^{-1}) \ll \log z$, and the remaining factors satisfy

$$\prod_{\substack{p\mid M\\ p>z}}(1+p^{-1}) \le \exp\left(\frac{\log M}{z\log z}\right) = O(1).$$

Thus $H_B(x) \gg \log x/\log z$. For the upper bound, discard $(n, M) = 1$ and use the finite Euler product:

$$\sum_{\substack{n\le x\\ n\text{ squarefree}\\ (n,P(z))=1}}\frac{1}{n} \le \prod_{z<p\le x}(1+p^{-1}) \ll_{b_0} \frac{\log X}{\log z}.$$ □

Lemma 3.5 (prime counts in progressions)**.** *Uniformly for $(a, q) = 1$ and $y \ge 2^{16}q$,*

$$\pi(y; q, a) \ll \frac{y}{\varphi(q)\log(y/q)}. \tag{8}$$

Proof. We use the block upper bound of Hooley's almost pure sieve [9], in the form of Ford–Halberstam [8, (2)–(3), (18)–(20)]. Put $w = (y/q)^{1/16}$ and $w_j = w^{2^{-j}}$. For $w > 2$, stop at the first J with $w_J \le 2$, and partition the primes $p < w$, $p \nmid q$, into the blocks

$$\mathcal{P}_j = \{p : \max\{2, w_{j+1}\} \le p < w_j,\ p \nmid q\}, \qquad 0 \le j < J.$$

For $w = 2$ the partition is empty. Write $P_j = \prod_{p\in\mathcal{P}_j} p$, $t_j = 2(j+1)$, and

$$S_j(n) = \sum_{\substack{d\mid(n,P_j)\\ \omega(d)\le t_j}}\mu(d).$$

An even inclusion–exclusion truncation is nonnegative and at least the indicator of $(n, P_j) = 1$: for $r > 0$, $\sum_{k=0}^{t}(-1)^k\binom{r}{k} = \binom{r-1}{t}$ when t is even, with zero binomial coefficients beyond their range. Thus

$$\mathbf{1}_{(n,\prod_j P_j)=1} \le \prod_j S_j(n).$$

Expand the product. Each term has a squarefree divisor $d = \prod_j d_j$, with $d_j \mid P_j$ and $\omega(d_j) \le t_j$. The disjoint blocks recover $d_j = (d, P_j)$, so every coefficient is 0 or $\mu(d)$. Also

$$d \le \prod_j w_j^{t_j} \le w^{\sum_{j\ge 0} 2(j+1)2^{-j}} = w^8.$$

Every d is coprime to q, and its multiples in the progression contribute $y/(qd) + O(1)$. Primes at most w contribute at most w. Consequently

$$\pi(y; q, a) \le \frac{y}{q}\prod_j U_j + O(w^8 + w), \qquad U_j = \sum_{\substack{d\mid P_j\\ \omega(d)\le t_j}} \frac{\mu(d)}{d}.$$

Put $H_j = \sum_{p\in\mathcal{P}_j} p^{-1}$ and $W_j = \prod_{p\in\mathcal{P}_j}(1 - p^{-1})$. Lemma 3.1 bounds H_j by an absolute B_0: each complete block lies between v and v^2, and the last block contains only primes below 4. Moreover $W_j^{-1} \le e^{2H_j} \le e^{2B_0}$. Bonferroni's inequality for the finite product gives

$$0 \le U_j - W_j \le \frac{H_j^{t_j+1}}{(t_j+1)!}, \qquad \prod_j \frac{U_j}{W_j} \le \exp\left(e^{2B_0}\sum_{j\ge 0}\frac{B_0^{2j+3}}{(2j+3)!}\right) \ll 1.$$

Hence, again by Lemma 3.1,

$$\frac{y}{q}\prod_j U_j \ll \frac{y}{q}\prod_{\substack{p<w\\ p\nmid q}}(1-p^{-1}) \le \frac{y}{\varphi(q)}\prod_{p<w}(1-p^{-1}) \ll \frac{y}{\varphi(q)\log(y/q)}.$$

The strict cutoff changes the Mertens product by at most one factor, bounded by 2. Finally $w^8 + w \ll (y/q)/\log(y/q)$ for $y/q \ge 2^{16}$, which is absorbed into the same bound since $\varphi(q) \le q$. □

Lemma 3.6 (reciprocal prime bound)**.** *Uniformly for $q \le X^\nu$ with $\nu < \beta$, and for reduced residue classes $a \pmod q$,*

$$\sum_{\substack{X^\beta < p \le X^{1-\sigma}\\ p\equiv a \pmod q}} \frac{1}{p} \ll_{\nu,\beta,\sigma} \frac{1}{\varphi(q)}.$$

Proof. Cover the prime window by $O(\log X)$ complete dyadic intervals $T < p \le 2T$ with $T \ge X^\beta/2$. Since $q \le X^\nu$ and $\nu < \beta$, we have $2T/q \ge 2^{16}$ for all sufficiently large X, uniformly in q. The bound (8) gives

$$\begin{aligned}\#\{T < p \le 2T : p \equiv a \pmod q\} &\le \#\{p \le 2T : p \equiv a \pmod q\}\\ &\ll \frac{T}{\varphi(q)\log(2T/q)} \ll_{\nu,\beta} \frac{T}{\varphi(q)\log X}.\end{aligned}$$

Multiplying by $1/T$ and summing proves the claim. □

Theorem 3.7 (Bombieri–Vinogradov)**.** *For every fixed $A > 0$ and $0 < \vartheta < 1/2$, uniformly for $x \ge 3$,*

$$\sum_{q\le x^\vartheta}\max_{(a,q)=1}\left|\pi(x;q,a) - \frac{\pi(x)}{\varphi(q)}\right| \ll_{A,\vartheta} \frac{x}{(\log x)^A}.$$

This is the classical theorem of Bombieri and Vinogradov [3, 17], in the $\pi(x)/\varphi(q)$ centering recorded in [10, Thm. 1.1]. The usual range $\sqrt{x}/(\log x)^B$ contains x^ϑ for sufficiently large x. In the $\mathrm{li}(x)/\varphi(q)$ form

of [13, Thm. 18.9, (18.11)], recentering follows from its $q = 1$ case and $\sum_{q\le Q} 1/\varphi(q) \ll \log(2Q)$, increasing the logarithmic saving first.

Lemma 3.8 (reciprocal primes in progressions)**.** *Put*

$$\mathcal{P}_X = \sum_{X^\beta < p \le X^{1-\sigma}} \frac{1}{p}, \qquad G_q(a) = \sum_{\substack{X^\beta < p \le X^{1-\sigma} \\ p \equiv a \pmod q}} \frac{1}{p} - \frac{\mathcal{P}_X}{\varphi(q)}.$$

For the exponents in (4)*,*

$$\sum_{q \le X^\nu} \tau(q) \max_{(a,q)=1} |G_q(a)| \ll (\log X)^{-7}.$$

Proof. Write $T_0 = X^\beta$, $T_1 = X^{1-\sigma}$, and $E(t; q, a) = \pi(t; q, a) - \pi(t)/\varphi(q)$. Since $X^\nu \le t^{\nu/\beta}$ for $t \ge T_0$ and $\nu/\beta = 12/25 < 1/2$, Theorem 3.7 with saving 20 gives

$$S(t) := \sum_{q \le X^\nu} \max_{(a,q)=1} |E(t; q, a)| \ll t(\log t)^{-20}.$$

Abel summation, taking the maximum inside the integral, gives

$$\sum_{q \le X^\nu} g_q \le \frac{S(T_1)}{T_1} + \frac{S(T_0)}{T_0} + \int_{T_0}^{T_1} \frac{S(t)}{t^2}\, dt \ll (\log X)^{-19}, \qquad g_q = \max_{(a,q)=1} |G_q(a)|.$$

Dropping primality and applying (7) gives $g_q \ll (\log X)/\varphi(q)$, since $q < X^\beta$ and $\mathcal{P}_X \ll 1$.

For $Q \ge 2$, expansion of the positive Euler product gives

$$\sum_{q \le Q} \frac{\tau(q)^2}{\varphi(q)} \le \prod_{p \le Q} \left(1 + \sum_{j \ge 1} \frac{(j+1)^2}{\varphi(p^j)}\right) = \prod_{p \le Q} \frac{1 + 3p^{-2} - 3p^{-3} + p^{-4}}{(1 - p^{-1})^4} \ll (\log(2Q))^4$$

by Lemma 3.1; the numerator product converges absolutely. Cauchy–Schwarz therefore yields

$$\sum_{q \le X^\nu} \tau(q) g_q \le \Big(\sum g_q\Big)^{1/2} \Big(\sum \tau(q)^2 g_q\Big)^{1/2} \ll (\log X)^{-19/2} (\log X)^{5/2} = (\log X)^{-7}. \qquad \square$$

4 Cubic divisor sums and pairwise coprimality

This section concerns the numerator 4. The proof of Theorem 1 uses Sections 2, 3, 5, and 6. For $a \ge 1$, set

$$\Lambda_{\mathrm{int}}(a) = \{h : h \mid a^2,\ h \notin \{1, a, a^2\}\}, \qquad \Lambda_<(a) = \{h : h \mid a^2,\ 1 < h < a\}.$$

Lemma 4.1 (a divisor identity)**.** *Writing $\tau_3(n)$ for the number of ordered triples of factors of n, we have*

$$\tau(n^2) = \sum_{k^2 \mid n} \mu(k) \tau_3(n/k^2).$$

Consequently,

$$\sum_{n \le L} \frac{\tau(n^2)}{n} = \frac{1}{6\zeta(2)} (\log L)^3 + O((\log L)^2).$$

Proof. Both sides of the identity are multiplicative. At $n = p^j$, it reads $\binom{j+2}{2} - \binom{j}{2} = 2j + 1$, with the second term omitted for $j < 2$.

For later use, fix M. Counting a residue class gives $\#\{n \le t : n \equiv r \pmod M\} = t/M + O(1)$, with an absolute error independent of r. Partial summation therefore gives, for $j = 0, 1, 2$,

$$\sum_{\substack{n \le T \\ n \equiv r \pmod M}} \frac{\{\log(T/n)\}^j}{n} = \frac{(\log T)^{j+1}}{(j+1)M} + O_M((1 + \log T)^j).$$

For $j = 0$ the numerator is interpreted as 1. In the triple sum below, first summing in z gives $M^{-1}\log(T/(xy)) + O_M(1)$. The total contribution of this error is $O_M((1+\log T)^2)$, since $\sum_{xy\le T}(xy)^{-1} \ll (1+\log T)^2$. Apply the displayed formula with $j = 1$ to the sum over y and then with $j = 2$ to the sum over x. This gives, uniformly over all residue triples,

$$\sum_{\substack{xyz\le T\\ x\equiv r_1,\, y\equiv r_2,\, z\equiv r_3 \pmod M}} \frac{1}{xyz} = \frac{(\log T)^3}{6M^3} + O_M((1+\log T)^2). \tag{9}$$

With $M = 1$, the divisor identity now yields

$$\sum_{n\le L}\frac{\tau(n^2)}{n} = \frac{1}{6}\sum_{k\le\sqrt{L}}\frac{\mu(k)}{k^2}\{\log(L/k^2)\}^3 + O((1+\log L)^2).$$

For $k \le \sqrt{L}$, the mean value theorem gives

$$\left|\{\log(L/k^2)\}^3 - (\log L)^3\right| \le 6(\log L)^2\log k.$$

Thus replacing the cubic logarithm costs

$$O((\log L)^2\sum_k(\log k)/k^2) = O((\log L)^2).$$

Completing the sum over k costs

$$O((\log L)^3\sum_{k>\sqrt{L}}k^{-2}) = O((\log L)^3/\sqrt{L}).$$

The completed sum is $\sum_k \mu(k)/k^2 = 1/\zeta(2)$. □

Proposition 4.2 (cubic divisor sum)**.** *As $L\to\infty$,*

$$\sum_{a\le L}\frac{|\Lambda_{\mathrm{int}}(a)|}{4a-1} \sim \frac{1}{24\zeta(2)}(\log L)^3.$$

Moreover,

$$\sum_{a\le L}\frac{|\Lambda_{<}(a)|}{4a-1} \sim \frac{1}{48\zeta(2)}(\log L)^3.$$

Proof. The involution $h\mapsto a^2/h$ pairs divisors below and above a. The excluded divisors $1, a, a^2$ contribute only $O(\log L)$ to the weighted sum. The preceding lemma also gives $\sum_{a\ge1}\tau(a^2)/a^2 < \infty$ by partial summation. Since

$$\frac{1}{4a-1} = \frac{1}{4a} + O(a^{-2}),$$

Lemma 4.1 gives the first asymptotic, and the asymptotic over the lower half is half of it. □

Proposition 4.3 (divisor sums with local restrictions)**.** *Let M be a fixed odd modulus and let $c \in (\mathbb{Z}/M\mathbb{Z})^\times$. Put*

$$a_M \equiv 4^{-1} \pmod{M}, \qquad u_c \equiv -c4^{-1} \pmod{M},$$

and define

$$\mathfrak{M}_{M,c}(L) = \sum_{\substack{a \le L \\ a \equiv a_M \pmod{M}}} \frac{1}{4a-1} \#\{h \in \Lambda_{\mathrm{int}}(a) : h \equiv u_c \pmod{M}\}.$$

Then, as $L \to \infty$,

$$\mathfrak{M}_{M,c}(L) = \frac{\mathfrak{S}(M)}{24\varphi(M)^2}(\log L)^3 + O_M((\log L)^2), \tag{10}$$

where

$$\mathfrak{S}(M) = \frac{1}{\zeta(2)} \prod_{\ell \mid M} \frac{(1-\ell^{-1})^3}{1-\ell^{-2}} > 0.$$

Coincident congruence classes are counted with their divisor multiplicities. For fixed M, the error term is uniform in the reduced class c. In particular, for each fixed odd prime ℓ, each fixed $k \ge 1$, and every $c \in (\mathbb{Z}/\ell^k\mathbb{Z})^\times$, the sum over the specified divisors compatible with the denominator class $n \equiv c \pmod{\ell^k}$ has positive cubic order. At the prime 2 there is a separate Chinese Remainder Theorem observation: because every modulus $4a-1$ is odd, an odd residue class $n \equiv c \pmod{2^k}$ can be imposed simultaneously with a congruence modulo $4a-1$. The 2-adic assertion concerns the denominator n.

Proof. If $a \equiv a_M \pmod{M}$, then $M \mid 4a-1$ and every divisor $h \mid a^2$ is coprime to $4a-1$. Thus $h \equiv u_c \pmod{M}$ makes the class $n \equiv -4h \pmod{4a-1}$ a reduced class contained in $n \equiv c \pmod{M}$.

Fix M throughout the proof and put

$$H_{M,c}(L) = \sum_{\substack{a \le L \\ a \equiv a_M \pmod{M}}} \frac{1}{a} \#\{h : h \mid a^2,\ h \equiv u_c \pmod{M}\}.$$

Thus all divisors are retained at this stage. The canonical decomposition in the proof of Theorem 2.2 gives $a = dfA$, $h = d^2 f$, $(d, A) = 1$, with each pair (a, h) counted exactly once. Expanding the last condition by Möbius inversion and writing $d = kx$, $A = ky$, we obtain

$$H_{M,c}(L) = \sum_{\substack{k \le \sqrt{L} \\ (k,M)=1}} \frac{\mu(k)}{k^2} \sum_{\substack{xyf \le L/k^2 \\ k^2xyf \equiv a_M,\ k^2x^2f \equiv u_c \pmod{M}}} \frac{1}{xyf}.$$

The first congruence forces k, x, y, f to be units modulo M. Dividing the second congruence by the first gives

$$xy^{-1} \equiv u_c a_M^{-1} \equiv -c \pmod{M}.$$

Conversely, for each of the $\varphi(M)$ unit classes of y, the choices $x \equiv -cy$ and $f \equiv a_M(k^2xy)^{-1}$ satisfy both congruences. Hence there are exactly $\varphi(M)$ residue triples, for every admissible k and every reduced c. By (9),

$$H_{M,c}(L) = \frac{\varphi(M)}{6M^3} \sum_{\substack{k \le \sqrt{L} \\ (k,M)=1}} \frac{\mu(k)}{k^2} \{\log(L/k^2)\}^3 + O_M((1+\log L)^2).$$

The error just displayed is bounded by

$$C_M \sum_{k \le \sqrt{L}} \frac{(1+\log(L/k^2))^2}{k^2} \ll_M (1+\log L)^2,$$

with the same C_M for every reduced c and every k. The bounds for replacing the logarithm and truncating the tail in Lemma 4.1 remain valid after restricting to $(k, M) = 1$. Since

$$\sum_{(k,M)=1} \frac{\mu(k)}{k^2} = \frac{1}{\zeta(2)} \prod_{\ell \mid M} (1 - \ell^{-2})^{-1},$$

the coefficient of $(\log L)^3$ is

$$\frac{\varphi(M)}{6M^3\zeta(2)} \prod_{\ell \mid M} (1 - \ell^{-2})^{-1} = \frac{1}{6\varphi(M)^2\zeta(2)} \prod_{\ell \mid M} \frac{(1 - \ell^{-1})^3}{1 - \ell^{-2}} = \frac{\mathfrak{S}(M)}{6\varphi(M)^2}.$$

Finally, replacing $1/a$ by $4/(4a - 1)$ costs $O(1)$, because $\sum_a \tau(a^2)/a^2 < \infty$, and removing the divisors $1, a, a^2$ costs $O_M(\log L)$. Restoring the factor $1/4$ proves (10), uniformly over the finitely many reduced classes. □

Lemma 4.4 (divisor sum over primes)**.** *For $L \geq 3$,*

$$\sum_{\substack{a \leq L \\ 4a-1 \text{ prime}}} \frac{\tau(a^2)}{a} \ll (\log L)^2.$$

Proof. Prime by prime, $\tau(n^2) \leq \tau_3(n)$, and every such triple has a factor at least $n^{1/3}$, so

$$\tau_3(n) \leq 3 \sum_{\substack{d \mid n \\ d \leq n^{2/3}}} \tau(d).$$

This hyperbola inequality is also used in [14, Lem. 4.1]; see [13, Ex. 20.2(a)]. The bound (8), applied with $y = 4x$, $q = 4d$ and the reduced class $-1 \pmod{4d}$, therefore gives

$$T(x) := \sum_{\substack{a \leq x \\ 4a-1 \text{ prime}}} \tau(a^2) \ll \sum_{d \leq x^{2/3}} \tau(d) \frac{x}{\varphi(4d) \log(x/d)} \ll \frac{x}{\log x} \sum_{d \leq x^{2/3}} \frac{\tau(d)}{\varphi(d)}.$$

For sufficiently large x, its range condition holds because $x/d \geq x^{1/3}$; the bounded remaining range is absorbed in the implied constant. Here $\log(x/d) \geq \frac{1}{3} \log x$. Using $n/\varphi(n) = \sum_{k \mid n} \mu^2(k)/\varphi(k)$ and $\tau(kt) \leq \tau(k)\tau(t)$, we have

$$\sum_{n \leq y} \frac{\tau(n)}{\varphi(n)} \leq (1 + \log y)^2 \sum_{k \geq 1} \frac{\mu^2(k)\tau(k)}{k\varphi(k)} = (1 + \log y)^2 \prod_p \left(1 + \frac{2}{p(p-1)}\right) \ll (1 + \log y)^2.$$

Thus $T(x) \ll x \log x$, and partial summation gives $T(L)/L + \int_1^L T(t)t^{-2}\, dt \ll (\log L)^2$, as required. □

Proposition 4.5 (pairwise coprime moduli)**.** *Let $\mathcal{B} \subset \{a \leq L\}$ be such that the integers $4a - 1$ with $a \in \mathcal{B}$ are pairwise coprime. Then*

$$\sum_{a \in \mathcal{B}} \frac{|\Lambda_{\mathrm{int}}(a)|}{4a - 1} \ll (\log L)^2.$$

Thus no subfamily of these congruences with pairwise coprime moduli can carry the cubic mass in Proposition 4.2.

Proof. Split according as $4a - 1$ is prime or composite. The prime part is bounded by Lemma 4.4, because $|\Lambda_{\mathrm{int}}(a)| \leq \tau(a^2)$ and $(4a - 1)^{-1} \ll a^{-1}$.

For the composite part, put $r = P^-(4a - 1)$. Pairwise coprimality makes these least prime factors distinct. Since $4a - 1$ is composite, $4a - 1 \geq r^2$. For every fixed $\eta_0 > 0$, $|\Lambda_{\mathrm{int}}(a)| \leq \tau(a^2) \ll_{\eta_0} a^{\eta_0}$ [13, Ex. 2.9(f)], and hence

$$\frac{|\Lambda_{\mathrm{int}}(a)|}{4a - 1} \ll_{\eta_0} r^{-2+2\eta_0}.$$

Choosing $\eta_0 < 1/4$ and summing over distinct r gives a bounded composite contribution. □

5 Counting compatible congruences

For each reduced class $b \pmod{P(z)}$, let I_b be a finite family of congruences with squarefree moduli $q_i \le X$, coprime to $P(z)$, and residues $c_i \pmod{q_i}$. Write $w_i = 1/q_i$ and $\mu_b = \sum_i w_i$. Two indices are compatible when their residues agree modulo (q_i, q_j); a set is compatible when every pair is. Put

$$L_S = \operatorname{lcm}(q_i : i \in S), \qquad L_\varnothing = 1, \qquad A(S) = \sum_{\substack{i \notin S \\ S \cup \{i\} \text{ compatible}}} w_i(q_i, L_S),$$

$$F_r = \sum_{\substack{S \subset I_b \\ |S| = r \\ S \text{ compatible}}} \frac{1}{L_S}, \qquad F_0 = 1.$$

Let Ω_b be the finite product probability space over all primes appearing in the moduli q_i. Explicitly,

$$\Omega_b = \prod_{\ell \mid \prod_{i \in I_b} q_i} \mathbb{Z}/\ell\mathbb{Z},$$

with independent uniform coordinates. The Chinese Remainder Theorem identifies each coordinate assignment at the primes dividing q_i with a residue modulo q_i. The event $\mathcal{A}_i^{(b)}$ is defined by the residue $c_i \pmod{q_i}$. Thus

$$\mathbb{P}(\mathcal{A}_i^{(b)}) = \frac{1}{q_i} = w_i^{(b)}.$$

If two events are compatible, the Chinese Remainder Theorem gives

$$\mathbb{P}(\mathcal{A}_i^{(b)} \cap \mathcal{A}_j^{(b)}) = \frac{1}{\operatorname{lcm}(q_i, q_j)} = \frac{(q_i, q_j)}{q_i q_j};$$

if they are incompatible, the intersection is empty. Two disjoint groups whose moduli are coprime across the groups depend on disjoint coordinates and are independent as families.

Put

$$W_b = \sum_{i \in I_b} \mathbf{1}_{\mathcal{A}_i^{(b)}}.$$

The Chinese Remainder Theorem gives $F_r = \mathbb{E}\binom{W_b}{r}$: an intersection of r compatible events has probability $1/L_S$, and an incompatible one is empty. For distinct events $i \ne j$, let $i \sim j$ denote $(q_i, q_j) > 1$, and define

$$\delta_b = \sup_i \sum_{\substack{j \sim i \\ j \ne i}} \mathbb{P}(\mathcal{A}_j^{(b)}), \qquad \Delta_b = \sum_{\substack{\{i,j\} : i \sim j \\ i \ne j}} \mathbb{P}(\mathcal{A}_i^{(b)} \cap \mathcal{A}_j^{(b)}).$$

Fix $K \ge 13$. In this section assume, uniformly in b, that $\mu_b \to \infty$ as $X \to \infty$ and

$$A(S) \le 2\mu_b \quad (|S| \le 2K\mu_b,\ S \text{ compatible}), \qquad \delta_b \le \frac{1}{4}, \qquad \Delta_b = o(\mu_b).$$

Lemma 5.1 (binomial moments)**.** *Uniformly for* $0 \le r \le 2K\mu_b$*,*

$$F_r \le \frac{(2\mu_b)^r}{r!}.$$

Proof. The Chinese Remainder Theorem gives $1/\operatorname{lcm}(L_S, q_i) = w_i(q_i, L_S)/L_S$. Counting the r choices of an element to delete from a compatible set of size r gives

$$rF_r = \sum_{\substack{S \text{ compatible} \\ |S| = r-1}} \frac{A(S)}{L_S} \le 2\mu_b F_{r-1}.$$

Induct from $F_0 = 1$. □

Theorem 5.2 (probability of no event)**.** *Uniformly in b,*

$$\mathbb{P}(W_b = 0) \le \exp\{-(1 - o(1))\mu_b\}.$$

Proof. We give a conditioning proof related to Spencer's argument in [11, Lemmas 1–2 and proof of Thm. 7, preprint version]. Fix b and abbreviate $\mathcal{A}_i^{(b)}$, μ_b, δ_b, Δ_b by A_i, μ, δ, Δ. Incompatible neighbours have zero intersection probability and remain adjacent.

Put $B_S = \bigcap_{j \in S} A_j^c$. If $\mathbb{P}(B_{I_b}) = 0$, there is nothing to prove; otherwise every conditioning event B_S has positive probability. We first show by induction on $|S|$ that

$$\mathbb{P}(A_i \mid B_S) \le 2w_i \qquad (i \notin S).$$

Write $\Gamma = \{j \in S : j \sim i\}$ and $T = S \setminus \Gamma$. If Γ is empty, independence gives equality with w_i. Otherwise $|T| < |S|$, so induction and the union bound give

$$\mathbb{P}(B_\Gamma \mid B_T) \ge 1 - \sum_{j \in \Gamma} \mathbb{P}(A_j \mid B_T) \ge 1 - 2\delta.$$

Since A_i is independent of B_T, division by this lower bound proves $\mathbb{P}(A_i \mid B_S) \le w_i/(1 - 2\delta) \le 2w_i$.

For a lower bound, the same partition gives

$$\mathbb{P}(A_i \mid B_S) \ge \mathbb{P}(A_i \cap B_\Gamma \mid B_T) \ge w_i - \sum_{j \in \Gamma} \mathbb{P}(A_i \cap A_j \mid B_T).$$

Fix $j \in \Gamma$, put $E = \{k \in T : k \sim j\}$, and write $T_0 = T \setminus E$. No vertex of T_0 is adjacent to either i or j, so the pair (A_i, A_j) is independent of B_{T_0}. The bound just proved gives $\mathbb{P}(B_E \mid B_{T_0}) \ge 1 - 2\delta$, and hence

$$\mathbb{P}(A_i \cap A_j \mid B_T) \le \frac{\mathbb{P}(A_i \cap A_j \mid B_{T_0})}{\mathbb{P}(B_E \mid B_{T_0})} \le \frac{\mathbb{P}(A_i \cap A_j)}{1 - 2\delta}.$$

Order the vertices and apply the lower bound with S equal to the preceding vertices, so that each unordered edge contributes once. The chain rule, $1 - t \le e^{-t}$, and these uniform bounds give

$$\mathbb{P}(W_b = 0) \le \exp\left\{-\mu + \frac{\Delta}{1 - 2\delta}\right\} \le \exp\{-\mu + 2\Delta\} = \exp\{-(1 - o(1))\mu_b\},$$

uniformly in b. □

For $r \ge 0$, let

$$C_r^{(b)}(N) = \sum_{\substack{S \subset I_b \\ |S| = r}} \#\{n \le N : n \equiv b \pmod{P(z)},\ n \in \mathcal{A}_i^{(b)} \text{ for all } i \in S\}.$$

Lemma 5.3 (approximation on a finite interval in one residue class)**.** *For every $r \ge 0$,*

$$C_r^{(b)}(N) = \frac{N}{P(z)} F_r + O(X^r F_r).$$

Proof. If S is compatible, the residual intersection is one class modulo $L_S = \operatorname{lcm}(q_i : i \in S)$. Since $(L_S, P(z)) = 1$, the condition $n \equiv b \pmod{P(z)}$ gives one class modulo $P(z)L_S$, contributing

$$\frac{N}{P(z)L_S} + O(1).$$

Incompatible intersections are empty. The number of compatible sets is at most $X^r F_r$, because $L_S \le X^r$ for every set of size r. Summing proves the claim. □

Theorem 5.4 (finite counting estimate)**.** *Fix $K \ge 13$ and put $R_b = \lfloor K\mu_b \rfloor$. Suppose that*

$$P(z) X^{2R_b} \exp\{2\mu_b\} \le N \exp\{-4\mu_b\}. \tag{11}$$

Then, uniformly in b,

$$\#\{n \le N : n \equiv b \pmod{P(z)},\quad n \notin \mathcal{A}_i^{(b)} \text{ for every } i \in I_b\} \le \frac{N}{P(z)} \exp\{-(1 - o(1))\mu_b\}.$$

Proof. Write $M_b(N)$ for the count in the conclusion and put $r = 2R_b$. Lemma 5.1 gives $F_s \le (2\mu_b)^s/s!$ for $s \le r$. For even r, the identity $\sum_{s=0}^{r}(-1)^s\binom{w}{s} = (-1)^r\binom{w-1}{r}$ for $w \ge 1$ gives

$$\mathbf{1}_{w=0} \le \sum_{s=0}^{r}(-1)^s\binom{w}{s} \le \mathbf{1}_{w=0} + \binom{w}{r}.$$

Apply the first inequality to integers in the class $b \pmod{P(z)}$ and the second in the Chinese Remainder Theorem model. Each compatible intersection has a counting error of absolute value at most 1, as in Lemma 5.3. Thus

$$\begin{aligned} M_b(N) &\le \frac{N}{P(z)}\{\mathbb{P}(W_b = 0) + F_r\} + \sum_{s=0}^{r} X^s F_s \\ &\le \frac{N}{P(z)}\left\{\mathbb{P}(W_b = 0) + \frac{(2\mu_b)^r}{r!}\right\} + X^r e^{2\mu_b}. \end{aligned}$$

For large X, $\mu_b \ge 2$ and $r \ge 25\mu_b$. The estimate $\log(r!) \ge \int_1^r \log t\, dt \ge r\log r - r$ implies

$$\frac{(2\mu_b)^r}{r!} \le (2e\mu_b/r)^r \le (2e/25)^r < 2^{-r} \le e^{-4\mu_b}.$$

Condition (11) bounds $X^r e^{2\mu_b}$ by $(N/P(z))e^{-4\mu_b}$. Theorem 5.2 estimates the remaining term and proves the result. □

Lemma 5.5 (lifting)**.** *Let $\mathcal{E}$ be a set of positive integers closed under taking divisors. Suppose $2 \le y \ll \log N$ and, uniformly for $\sqrt{N} \le T \le N$,*

$$\#\{u \le T : u \in \mathcal{E},\ (u, P(y)) = 1\} \le ATe^{-F}$$

for $A > 0$ and $F \ge 0$. Then

$$\#(\mathcal{E} \cap [1, N]) \ll AN\log(2y)e^{-F} + N^{3/4+o(1)}.$$

The error is uniform for a fixed constant in $y \ll \log N$.

Proof. Write $n = su$ with s supported on primes at most y and $(u, P(y)) = 1$. The contribution with $s > \sqrt{N}$, even without requiring $n \in \mathcal{E}$, is at most

$$N \sum_{\substack{s>\sqrt{N} \\ P^+(s)\le y}} \frac{1}{s} \le N^{3/4}\prod_{p\le y}(1 - p^{-1/2})^{-1} = N^{3/4+o(1)}.$$

Indeed, the logarithm of the product is $O(\sum_{2\le n\le y} n^{-1/2}) = O(\sqrt{y}) = O(\sqrt{\log N})$, uniformly under the stated bound on y. For $s \le \sqrt{N}$, put $T = N/s \in [\sqrt{N}, N]$. Since $u \mid n$, divisor closure gives $u \in \mathcal{E}$. Summing the hypothesis over s bounds this contribution by $ANe^{-F}\prod_{p\le y}(1 - p^{-1})^{-1} \ll ANe^{-F}\log(2y)$. □

6 The selected family

Fix $m \ge 4$ for the rest of the proof; all thresholds may depend on m. We take X large enough that $z > m$ and put

$$S_0 = X^\eta, \qquad Y_0 = X^\lambda, \qquad Y = X^\sigma, \qquad H = X^\theta.$$

Define $\mathcal{F}$ to be the set of integers

$$e = rs^2, \qquad \mathrm{rad}(e) = rs,$$

where r and s are squarefree, $(r, s) = 1$, $(rs, m) = 1$, and

$$1 \le s \le S_0, \qquad \frac{Y_0}{s} < r \le \frac{H}{s^2}. \tag{12}$$

Here $\mathrm{rad}(e)$ is the product of the distinct prime factors of e. The squarefree, coprime coordinates r, s are uniquely determined by $e = rs^2$. In the application e is a divisor of a^2, as in Section 2. The ranges give $e \le H$ and

$$\mathrm{rad}(e) = rs > Y_0 > Y. \tag{13}$$

We first fix a reduced residue class modulo $P(z)$:

$$b \in (\mathbb{Z}/P(z)\mathbb{Z})^\times.$$

For $e \in \mathcal{F}$, choose a small squarefree odd divisor d_- and a rough squarefree cofactor d_+ such that

$$d_- \le U, \qquad d_- \mid P(z), \qquad b \equiv -me \pmod{d_-}, \tag{14}$$

$$d_-d_+ \le Y, \qquad (d_+, P(z)) = 1, \qquad (d_-d_+, m\,\mathrm{rad}(e)) = 1. \tag{15}$$

Finally choose a prime p satisfying

$$X^\beta < p \le X^{1-\sigma}, \qquad d_-d_+p \equiv -1 \pmod{m\,\mathrm{rad}(e)}. \tag{16}$$

For every quadruple $i = (e, d_-, d_+, p)$ satisfying these conditions, define

$$Q_i = d_-d_+p, \qquad q_i = d_+p,$$

and

$$a_i = \frac{Q_i + 1}{m}, \qquad \mathcal{A}_i^{(b)} = \{n : n \equiv -me \pmod{q_i}\}, \qquad w_i^{(b)} = \frac{1}{q_i}.$$

The full modulus is Q_i; after conditioning on $n \equiv b \pmod{P(z)}$, only the residual modulus q_i contributes to the density. Indeed, $d_- \mid P(z)$ and (14) give $n \equiv -me \pmod{d_-}$. Combining this with $\mathcal{A}_i^{(b)}$ gives

$$n \equiv -me \pmod{Q_i}.$$

The congruence (16) implies that a_i is an integer and $\mathrm{rad}(e) \mid a_i$. Because $e = rs^2$ and $\mathrm{rad}(e) = rs$, this gives $e \mid a_i^2$; also $Q_i = ma_i - 1$. Lemma 2.3 therefore gives a representation. Let I_b be the set of all quadruples above and set

$$\mu_b = \sum_{i \in I_b} w_i^{(b)}. \tag{17}$$

Since $d_-d_+ \le X^\sigma$ and $p \le X^{1-\sigma}$, both q_i and Q_i are at most X.

We record a bound for reciprocal sums used in Lemma 6.1 and Theorem 6.3. For any $0 < u < v$, the count on $(u, t]$ in a class modulo q is $(t-u)/q + O(1)$. Partial summation therefore gives the reciprocal sum in that class as $q^{-1}\log(v/u) + O(1/u)$, whether or not the class is reduced. If M is squarefree and $(M, d) = 1$, finite inclusion–exclusion over the divisors of M consequently gives

$$\sum_{\substack{u < n \le v \\ n \equiv a \pmod d \\ (n,M)=1}} \frac{1}{n} = \frac{\varphi(M)}{Md}\log\frac{v}{u} + O\left(\frac{\tau(M)}{u}\right).$$

This formula is uniform in the modulus and residue, including when $u < 1$.

For arbitrary positive integers d, M, with M squarefree and $(M, d) = 1$, every reduced class $a \pmod d$, and $1 \le A < B$, expand $1/\varphi(r) = r^{-1}\sum_{k \mid r}\gamma(k)$ with nonnegative coefficients $\gamma(k) = \mu^2(k)/\varphi(k)$. Only $(k, dM) = 1$ occurs. Writing $r = kn$ and applying the preceding formula with endpoints $A/k, B/k$ gives

$$\begin{aligned}\sum_{\substack{A < r \le B \\ r \equiv a \pmod d \\ (r,M)=1}} \frac{1}{\varphi(r)} &= \frac{\varphi(M)}{Md}\log\frac{B}{A}\sum_{\substack{k \le B \\ (k, dM)=1}}\frac{\gamma(k)}{k} + O\left(\frac{\tau(M)}{A}\sum_{k \le B}\gamma(k)\right) \\ &\le C_\gamma\frac{\varphi(M)}{Md}\log\frac{B}{A} + O\left(\frac{\tau(M)\log(2B)}{A}\right).\end{aligned} \tag{18}$$

The error in the inner sum is $O(\tau(M)k/A)$. Multiplication by $\gamma(k)/k$ and (5) give the displayed error bound.

Lemma 6.1 (average of $1/\varphi$ in progressions)**.** *Let $d \le U$ be squarefree and odd, let $(a,d)=1$, and let $s \le S_0$ be squarefree with $(s,m)=(sm,d)=1$. Put $U_0 = Y_0/s$ and $U_1 = H/s^2$. Uniformly in all these parameters,*

$$\sum_{\substack{U_0 < r \le U_1 \\ r \text{ squarefree} \\ (r,sm)=1 \\ r \equiv a \pmod d}} \frac{1}{\varphi(r)} \asymp_m \frac{1}{d}\frac{\varphi(s)}{s}\log\frac{U_1}{U_0}.$$

Proof. Now take $A = Y_0/s$, $B = H/s^2$, $L_s = \log(B/A)$, and $M = s\,\mathrm{rad}(m)$. The restrictions on s, d make M squarefree and $(M,d)=1$. For the lower bound, $1/\varphi(r) \ge 1/r$, and the Möbius calculation in the proof of Lemma 3.2 gives

$$\sum_{\substack{A < r \le B \\ r \equiv a \pmod d \\ (r,sm)=1}} \frac{\mu^2(r)}{r} = \kappa(M,d)\frac{\varphi(M)}{Md}L_s + O_m(\tau(s)/\sqrt{A}), \qquad \frac{\varphi(M)}{M} \asymp_m \frac{\varphi(s)}{s}.$$

For the upper bound, discard squarefreeness and apply (18) with this M. Since s is squarefree,

$$\frac{\tau(s)}{\varphi(s)} = \prod_{p \mid s}\frac{2}{p-1} \le 2.$$

As $L_s \asymp \log X$, the two errors divided by $\varphi(s)L_s/(sd)$ are

$$O_m\left(\frac{dX^{(3\eta-\lambda)/2}}{\log X}\right) \quad \text{and} \quad O_m\left(dX^{2\eta-\lambda}\right).$$

Both tend uniformly to zero for $d \le U = (\log X)^8$, since $3\eta < \lambda$. The constants and threshold are independent of a, d, s. □

Proposition 6.2 (quadratic divisor sums)**.** *The selected divisor family, before any sieving at small primes, satisfies*

$$M_1 := \sum_{e \in \mathcal{F}} \frac{1}{\mathrm{rad}(e)} \asymp_m (\log X)^2, \qquad M_\varphi := \sum_{e \in \mathcal{F}} \frac{1}{\varphi(\mathrm{rad}(e))} \asymp_m (\log X)^2.$$

Proof. Write $e = rs^2$ and put $L_s = \log(H/(Y_0 s))$. For all $s \le S_0$,

$$(\theta - \lambda - \eta)\log X \le L_s \le (\theta - \lambda)\log X.$$

Lemma 3.2 and Lemma 6.1 with $d=1$ give, respectively, the sums over r of $1/r$ and $1/\varphi(r)$ as order $(\varphi(s)/s)L_s$, uniformly in s, with coprimality modulus $s\,\mathrm{rad}(m)$. Since $(r,s)=1$ on the family, $\varphi(rs)=\varphi(r)\varphi(s)$. Hence

$$M_1 \asymp_m \log X \sum_{\substack{s \le S_0 \\ (s,m)=1}} \frac{\mu^2(s)\varphi(s)}{s^2}, \qquad M_\varphi \asymp_m \log X \sum_{\substack{s \le S_0 \\ (s,m)=1}} \frac{\mu^2(s)}{s}.$$

Lemma 3.3 completes both estimates. □

Theorem 6.3 (divisor sums in residue classes)**.** *Let $D \le YU$ be squarefree and let $(c,D)=1$. Then*

$$\sum_{\substack{e \in \mathcal{F} \\ e \equiv c \pmod D}} \frac{1}{\mathrm{rad}(e)} \ll \frac{M_1}{D}, \qquad \sum_{\substack{e \in \mathcal{F} \\ e \equiv c \pmod D}} \frac{1}{\varphi(\mathrm{rad}(e))} \ll \frac{M_\varphi}{D}.$$

Proof. Write $e = rs^2$. Terms with $(s, D) > 1$ vanish, and otherwise $e \equiv c \pmod D$ is the reduced progression $r \equiv cs^{-2} \pmod D$. Since $(r, s) = 1$ on the family, $\varphi(\mathrm{rad}(e)) = \varphi(r)\varphi(s)$. Discard the squarefree and coprimality restrictions on r for an upper bound, and use (18) with $M = 1$, $d = D$, $A = Y_0/s$, and $B = H/s^2$. Then

$$\sum_{\substack{e\in\mathcal{F}\\ e\equiv c \pmod D}} \frac{1}{\varphi(\mathrm{rad}(e))} \ll \frac{\log X}{D}\sum_{s\le S_0}\frac{1}{\varphi(s)} + \frac{\log X}{Y_0}\sum_{s\le S_0}\frac{s}{\varphi(s)} \ll \frac{(\log X)^2}{D} + \frac{S_0\log X}{Y_0}.$$

The last step is (5). Relative to the first term, the endpoint error is at most

$$\frac{DS_0}{Y_0\log X} \le X^{\eta+\sigma-\lambda}(\log X)^7 = o(1),$$

uniformly for $D \le YU$. Proposition 6.2 proves the second assertion. The first follows from $1/\mathrm{rad}(e) \le 1/\varphi(\mathrm{rad}(e))$ and $M_\varphi \asymp M_1$. □

Lemma 6.4 (averaged sum over divisors with small prime factors)**.** *For*

$$A_s := \sum_{\substack{d\le U\\ d\mid P(z)\\ d \text{ odd}\\ (d,sm)=1}} \frac{1}{d}$$

one has

$$\sum_{\substack{s\le S_0\\ s \text{ squarefree}\\ (s,m)=1}} \frac{A_s}{s} \asymp_m \log S_0 \log z.$$

Proof. Since $U = z^2$, the upper bound follows from $A_s \le \sum_{d\le U} d^{-1} \le 1 + 2\log z$ and Lemma 3.3.

For the lower bound, put

$$J_d(T) = \sum_{\substack{s\le T\\ (s,dm)=1}} \frac{\mu^2(s)}{s}.$$

For squarefree d coprime to m, each squarefree s coprime to m has a unique decomposition $s = gt$, where $g = (s, d) \mid d$ and $(t, d) = 1$. Consequently

$$J_1(S_0) = \sum_{g\mid d}\frac{1}{g}J_d(S_0/g) \le J_d(S_0)\prod_{p\mid d}(1+p^{-1}) \le \frac{d}{\varphi(d)}J_d(S_0).$$

Thus $J_d(S_0) \ge (\varphi(d)/d)J_1(S_0)$ for every squarefree d coprime to m. Retain odd squarefree $d \le z$ with $(d, m) = 1$, which divide $P(z)$ and satisfy $d \le U$. Interchanging the positive sums gives

$$\sum_{\substack{s\le S_0\\ (s,m)=1}} \frac{\mu^2(s)A_s}{s} \ge \sum_{\substack{d\le z\\ d \text{ odd},\ (d,m)=1}} \frac{\mu^2(d)}{d}J_d(S_0) \ge J_1(S_0) \sum_{\substack{d\le z\\ d \text{ odd},\ (d,m)=1}} \frac{\mu^2(d)\varphi(d)}{d^2} \gg \log S_0 \log z.$$

The last bounds are the ordinary squarefree reciprocal sum and the odd weighted sum proved in Lemma 3.3, applied with the fixed restrictions $(s, m) = 1$ and $(d, 2m) = 1$. □

Lemma 6.5 (average over small divisors)**.** *Uniformly for every $b \in (\mathbb{Z}/P(z)\mathbb{Z})^\times$,*

$$\Sigma_b := \sum_{\substack{d_-\le U\\ d_-\mid P(z)\\ d_- \text{ odd}}} \sum_{\substack{e\in\mathcal{F}\\ b\equiv -me \pmod{d_-}}} \frac{1}{\varphi(m\,\mathrm{rad}(e))} \asymp_m (\log X)^2\log z.$$

Proof. Write $e = rs^2$. For fixed s, observe that a divisor d_- contributes only when $(d_-, sm) = 1$, in which case the congruence is the reduced linear progression $r \equiv -b(ms^2)^{-1} \pmod{d_-}$. Since $(rs, m) = 1$ and $(r, s) = 1$, $\varphi(mrs) = \varphi(m)\varphi(r)\varphi(s)$. By Lemma 6.1,

$$\sum_{\substack{Y_0/s<r\le H/s^2\\ r \text{ squarefree},\ (r,sm)=1\\ b\equiv -mrs^2 \pmod{d_-}}} \frac{1}{\varphi(mrs)} \asymp_m \frac{1}{sd_-}\log\frac{H}{Y_0 s}.$$

Since $\log(H/(Y_0 s)) \asymp \log X$ for all $s \le S_0$, summing over d_- and using Lemma 6.4 we obtain

$$\Sigma_b \asymp \log X \sum_{\substack{s\le S_0\\ s \text{ squarefree}\\ (s,m)=1}} \frac{A_s}{s} \asymp \log X \log S_0 \log z \asymp (\log X)^2 \log z.$$

Every comparison has a threshold independent of b, because b enters only through a reduced residue class modulo d_-. □

Proposition 6.6 (total event weight)**.** *Uniformly in* $b \in (\mathbb{Z}/P(z)\mathbb{Z})^\times$,

$$\mu_b \asymp_m (\log X)^3.$$

Proof. Use $\mathcal{P}_X$ and $G_q(a)$ from Lemma 3.8. Lemma 3.1 gives

$$\mathcal{P}_X = \log\frac{1-\sigma}{\beta} + o(1) > 0. \tag{19}$$

For every admissible (e, d_-, d_+) the prime condition is the reduced class $a = -(d_- d_+)^{-1} \pmod{m\,\mathrm{rad}(e)}$. This class is reduced by (15).

Define

$$V(e, d_-) = \sum_{\substack{d_+\le Y/d_-\\ d_+ \text{ squarefree}\\ (d_+,P(z))=(d_+,m\,\mathrm{rad}(e))=1}} \frac{1}{d_+}.$$

Substituting the discrepancy into (17) gives

$$\mu_b = \mathcal{P}_X \mathcal{M}_b + \mathcal{G}_b, \tag{20}$$

where

$$\mathcal{M}_b = \sum_{\substack{d_-\le U\\ d_-\mid P(z)\\ d_- \text{ odd}}} \sum_{\substack{e\in\mathcal{F}\\ b\equiv -me \pmod{d_-}}} \frac{V(e, d_-)}{\varphi(m\,\mathrm{rad}(e))} \tag{21}$$

and $\mathcal{G}_b = \sum_{(e,d_-,d_+)} d_+^{-1} G_{m\,\mathrm{rad}(e)}(a)$, summed over the admissible triples.

For large X, $X^{\sigma/2} \le Y/d_- \le X^\sigma$ for every $d_- \le U$. Lemma 3.4, applied with endpoint Y/d_- and auxiliary modulus $m\,\mathrm{rad}(e) \le X^{\theta+1}$, gives $V(e, d_-) \asymp \log X/\log z$ uniformly in e, d_- and b. Lemma 6.5 now gives $\mathcal{M}_b \asymp (\log X/\log z)\Sigma_b \asymp (\log X)^3$.

The exponents satisfy $\theta < \nu < \beta/2$. Since m is fixed, $mX^\theta \le X^\nu$ beyond a threshold depending on m. Thus every modulus $m\,\mathrm{rad}(e)$ is in the range of Lemma 3.8, with a fixed margin at the lower prime endpoint X^β.

For a fixed radical $t = \mathrm{rad}(e) = rs$, the divisor $s \mid t$ determines $e = ts$, so there are at most $\tau(t)$ choices of e. For each such e there are at most U choices of d_-, and, by enlarging the range to $d_+ \le Y$ and discarding the extra coprimality restrictions, Lemma 3.4 gives

$$\sum_{\text{admissible } d_+} \frac{1}{d_+} \le \sum_{\substack{d_+\le Y\\ d_+ \text{ squarefree}\\ (d_+,P(z))=1}} \frac{1}{d_+} \ll \frac{\log X}{\log z}.$$

Taking the maximum discrepancy over reduced residues before summing these multiplicities, and then using $\tau(t) \le \tau(mt)$ and $q = mt$, gives

$$|\mathcal{G}_b| \ll \frac{U \log X}{\log z} \sum_{q \le X^\nu} \tau(q) \max_{(a,q)=1} |G_q(a)| \ll_m \frac{(\log X)^2}{\log z} = o((\log X)^3). \tag{22}$$

The second bound is Lemma 3.8, since $U = (\log X)^8$. The maximum over reduced residues makes this error bound independent of b.

Uniformly in b, (22) gives

$$\frac{|\mathcal{G}_b|}{\mathcal{M}_b} \ll_m \frac{1}{\log X \log z} = o_m(1).$$

Equations (19) and (20) therefore imply

$$\frac{\mu_b}{\mathcal{M}_b} = \log \frac{1-\sigma}{\beta} + o_m(1) > 0.$$

Together with $\mathcal{M}_b \asymp_m (\log X)^3$, this proves the proposition. □

Proposition 6.7 (event weights in residue classes)**.** *Let $b \in (\mathbb{Z}/P(z)\mathbb{Z})^\times$. Let $D \le Y$ be squarefree with $(D, P(z)) = 1$, and let $(c, D) = 1$. Define*

$$B_{D,c}^{(b)} = \sum_{\substack{i=(e,d_-,d_+,p)\in I_b \\ D \mid d_+ \\ e \equiv c \pmod D}} w_i^{(b)}.$$

Then, uniformly in b, D, c in these ranges,

$$B_{D,c}^{(b)} \ll_m \frac{\mu_b}{D^2}.$$

Moreover,

$$\sum_{\substack{i=(e,d_-,d_+,p)\in I_b \\ D \mid d_+}} w_i^{(b)} \ll_m \frac{\mu_b}{D}.$$

Proof. Since $m \operatorname{rad}(e) \le m X^\theta \le X^\nu$ for large X, Lemma 3.6 applies uniformly and, for fixed e, d_-, d_+, gives $\sum_p w_i^{(b)} \ll_m 1/(d_+ \varphi(m \operatorname{rad}(e)))$, where the sum is over the admissible primes. Since $z > m$, also $(D, m) = 1$. If $D \mid d_+$, write $d_+ = Dt$; the coprimality conditions then require $(D, \operatorname{rad}(e)) = 1$. Dropping restrictions on t except squarefreeness and roughness, Lemma 3.4 gives

$$\sum_{\substack{d_+ \le Y/d_- \\ D \mid d_+ \\ d_+ \text{ admissible}}} \frac{1}{d_+} \le \frac{1}{D} \sum_{\substack{t \le Y \\ t \text{ squarefree} \\ (t, P(z))=1}} \frac{1}{t} \ll_m \frac{\log X}{D \log z}.$$

Contributing d_- satisfy $(d_-, m) = 1$ by (14), since b is reduced modulo $P(z)$. As $(D, d_-) = 1$, the conditions $e \equiv c \pmod D$ and $e \equiv -bm^{-1} \pmod{d_-}$ combine into a reduced class modulo $Dd_- \le YU$. Theorem 6.3 and $\varphi(m \operatorname{rad}(e)) = \varphi(m)\varphi(\operatorname{rad}(e))$ therefore give

$$\sum_{\substack{e \in \mathcal{F} \\ e \equiv c \pmod D \\ b \equiv -me \pmod{d_-}}} \frac{1}{\varphi(m \operatorname{rad}(e))} \ll_m \frac{M_\varphi}{D d_-}.$$

Consequently, since $\sum_{d \le U} 1/d \ll_m \log U \le 2 \log z$, Proposition 6.6 gives

$$B_{D,c}^{(b)} \ll_m \frac{M_\varphi \log X}{D^2 \log z} \sum_{\substack{d_- \le U \\ d_- \mid P(z)}} \frac{1}{d_-} \ll_m \frac{M_\varphi \log X}{D^2} \ll_m \frac{\mu_b}{D^2}.$$

For the second assertion, omit the congruence modulo D and use Theorem 6.3 only modulo d_-. The first factor D^{-1} remains, whereas the divisor sum is now $O(M_\varphi/d_-)$; the same calculation gives $O(\mu_b/D)$. □

Lemma 6.8 (uniqueness when the large prime is shared)**.** *Two distinct compatible events in I_b cannot share the same large prime p. For any prime $p > X^\beta$, the total weight of events having this large prime is at most $H/p \le X^{\theta-\beta} = o(1)$.*

Proof. Suppose (e, d_-, d_+, p) and (f, d'_-, d'_+, p) are compatible. Compatibility at p gives $me \equiv mf \pmod p$. For large X we have $p > m$, so $e \equiv f \pmod p$. Since $e, f \le H = X^\theta < X^\beta < p$, we have $e = f$. The defining congruences then give

$$d_-d_+p \equiv d'_-d'_+p \equiv -1 \pmod{m\,\mathrm{rad}(e)}.$$

Because $p > X^\beta > X^\theta \ge \mathrm{rad}(e)$, we have $(p, m\,\mathrm{rad}(e)) = 1$, so

$$d_-d_+ \equiv d'_-d'_+ \pmod{m\,\mathrm{rad}(e)}.$$

The products d_-d_+ and $d'_-d'_+$ are at most $Y < Y_0 < \mathrm{rad}(e)$, so they are equal. Since $d_- = (d_-d_+, P(z))$, their factorizations agree and the two events are identical.

For the direct count, events with the same e, p are compatible, hence identical by what was just proved. There are at most $\lfloor H \rfloor$ possible positive e, and each event has weight at most $1/p$. Summing gives the stated bound. □

Lemma 6.9 (overlap bounds)**.** *Uniformly in b,*

$$\delta_b = o_m(1), \qquad \Delta_b = o_m(\mu_b).$$

For every fixed $K > 0$ and every compatible S with $|S| \le K\mu_b$,

$$A(S) \le \mu_b(1 + o_{m,K}(1)).$$

Proof. The mass of neighbours sharing a medium prime is, by Proposition 6.7, at most

$$C_m\mu_b \sum_{\ell \mid d_{+,i}} \frac{1}{\ell} \le C_m\mu_b \frac{\log Y}{z \log z} \ll_m \frac{1}{\log\log X} = o_m(1).$$

Together with the direct count in Lemma 6.8, this proves the bound for δ_b.

For a distinct compatible dependent pair, the common primes are all medium primes, by uniqueness and $Y < X^\beta$. If $g = (q_i, q_j) > 1$, use $g \le 2\sum_{D \mid g,\ D>1} \varphi(D)$. Since every prime in D exceeds $z > m$, compatibility forces e_i, e_j into the same reduced class modulo D. Thus

$$\Delta_b \ll \sum_{\substack{1<D\le Y \\ D \text{ squarefree} \\ (D,P(z))=1}} \varphi(D) \sum_c (B^{(b)}_{D,c})^2 \ll_m \mu_b^2 \sum_{n>z} n^{-2} \ll_m \frac{\mu_b^2}{z} = o_m(\mu_b).$$

The middle step uses $\max_c B^{(b)}_{D,c} \ll_m \mu_b/D^2$ and $\sum_c B^{(b)}_{D,c} \ll_m \mu_b/D$.

For a compatible set S, let $J(S)$ be the primes dividing $\prod_{j\in S} d_{+,j}$. A compatible extension shares no large prime with S, so $(q_i, L_S) \mid d_{+,i}$. For each $\ell \in J(S)$, compatibility makes the e_j with $\ell \mid d_{+,j}$ congruent modulo ℓ; their common class is reduced because $(d_{+,j}, \mathrm{rad}(e_j)) = 1$. Thus every divisor D of this gcd prescribes one reduced class of e_i modulo D. Expanding $g = \sum_{D\mid g} \varphi(D)$, applying Proposition 6.7, and extending a positive divisor sum beyond $D \le Y$ give

$$\begin{aligned} A(S) &\le \mu_b + C_m\mu_b \sum_{\substack{D \mid \prod_{\ell\in J(S)} \ell \\ D>1}} \frac{\varphi(D)}{D^2} \\ &\le \mu_b \left[1 + C_m \left\{ \prod_{\ell\in J(S)} (1+\ell^{-1}) - 1 \right\}\right]. \end{aligned}$$

But

$$\sum_{\ell\in J(S)}\frac{1}{\ell}\le\frac{|S|\log Y}{z\log z}\le\frac{K\mu_b\log Y}{z\log z}\ll_{m,K}\frac{1}{\log\log X}=o_{m,K}(1).$$

The product is consequently $1+o_{m,K}(1)$, uniformly in b and S. □

6.1 Completion of the proof

Proof of Theorem 2. Fix $m\ge 4$ and $K=13$. Proposition 6.6 and Lemma 6.9, using the latter with parameter $2K$, verify the hypotheses of Section 5 for all sufficiently large X, uniformly in b. In particular $c_m(\log X)^3\le\mu_b\le C_m(\log X)^3$. Choose

$$\log X=\alpha_m(\log N)^{1/4},\tag{23}$$

with $\alpha_m>0$ sufficiently small. Lemma 3.1 gives $\log P(z)\ll z=(\log X)^4$. For $R_b=\lfloor K\mu_b\rfloor$,

$$\log P(z)+2R_b\log X+6\mu_b\ll_m(\log X)^4.$$

Taking α_m small makes this at most $\frac12\log N$. Since $\log T\ge\frac12\log N$ for $\sqrt{N}\le T\le N$, this choice of X, cutoff, and family satisfies (11) throughout that range. Theorem 5.4, summed over the reduced classes modulo $P(z)$, gives

$$\#\{n\le T:(n,P(z))=1,\ m/n\text{ has no representation}\}\ll_m T\exp\{-c'_m(\log N)^{3/4}\}.$$

The exceptional set is closed under taking divisors: a representation for a divisor of n scales to one for n. Apply Lemma 5.5 at the common cutoff $z=\alpha_m^4\log N$. The factor $\log(2z)$ and the error $N^{3/4+o_m(1)}$ are absorbed by decreasing c'_m. Enlarging the implied constant covers the finite remaining range of N. □

Proof of Theorem 1. Take $m=4$ in Theorem 2 and use $E(N)\le E_4(N)$. The assertion for prime denominators follows as well, since $\pi(N)\gg N/\log N$ and the factor $\log N$ is absorbed by decreasing the constant in the exponential. □

7 Further remarks

Remark 7.1 (geometry). For numerator 4, the surface $U_n:4xyz=n(xy+xz+yz)$ is an affine part of a Cayley cubic; see [4]. In projective coordinates $(x_0:x_1:x_2:x_3)$, its closure is

$$4x_1x_2x_3=nx_0(x_1x_2+x_1x_3+x_2x_3).$$

Fixing $x=an$ gives $(4a-1)yz=an(y+z)$, so the projective closures of these curves are the conic components of explicit plane sections. On the plane $x_1=anx_0$, the cubic factors as

$$nx_0\big((4a-1)x_2x_3-anx_0(x_2+x_3)\big)=0.$$

The factor x_0 gives the boundary line $x_0=x_1=0$.

Remark 7.2 (effectivity). The thresholds obtained here inherit the ineffectivity of the usual Bombieri–Vinogradov theorem. Johnston [10] gives effective variants for sifting problems; applying them to this family requires a separate analysis.

Use of artificial intelligence. MIT CSAIL Mantis was used for Lean 4 autoformalization.
Data and code availability. The Lean source is available in the accompanying repository [12].

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